



MRS6241-P2840-M81 mmWave Radar Module Datasheet

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1 Overview

1.1 Module Introduction

MRS6241-P2840-M81 is a high-performance, single-chip 60 GHz 2T4R FMCW MIMO radar sensor. The device consists of three subsystems:

- **mmWave radar subsystem:** This subsystem includes all high-performance radar RF, analog, and baseband circuitry. It also includes the chirp generator and the TX-to-RX operating state machine. The hardware accelerator (HWA) is integrated in the baseband to offload specified radar-processing tasks such as 1D/2D FFT and CFAR.
- **Application subsystem:** This subsystem primarily includes two 32-bit RISC-V CPU cores with a DSP, FPU, and instruction/data (I/D) caches. The CPUs connect to SRAM, peripheral interfaces, and other subsystems through the AHB and APB buses. The RISC-V CPUs run radar post-processing algorithms, wireless protocols, and other system tasks.
- **Power and clock subsystem:** The chip is a complex single-chip SoC with multiple functional systems that are centrally managed by one power and clock subsystem. This subsystem includes multiple clock sources (including LSI, MSI, DCXO, and SPLL), power modules (LDO, BG, and clamp circuits), and a temperature sensor.

1.2 Module Specifications

Table 1-1 Module Specifications

Specification	Min.	Typ.	Max.	Unit	Remarks
Radar Specifications					
Operating Frequency Band	60.0	-	62.0	GHz	
Continuous Sweep Bandwidth	-	2.0	-	GHz	
TX/RX Channels	2T4R			-	
Equivalent Isotropically Radiated Power (EIRP)	-	23.7	-	dBm	
Horizontal FOV	-	160	-	°	
Vertical FOV	-	30	-	°	
Maximum Detection Range	-	-	-	m	Depends on the radar cross section (RCS) of the target
Minimum Detection Range	-	-	0.01	m	Based on a human target
Hardware Specifications					
Module Dimensions	-	28 × 40	-	mm ²	
VCC Supply Voltage	1.7	3.3	5.5	V	
VDDIO Supply	1.7	3.3	3.6	V	

Voltage					
Operating Temperature	-40	-	85	°C	Ambient temperature

1.3 Applications

The MRS6241-P2840-M81 module can be widely used in various AIoT smart-sensing applications, such as drones, two-wheelers, and people detection, providing highly accurate and sensitive detection of target motion and micro-motion.

- Occupancy Detection
- Human Tracking/Counting
- Industrial Inspection
- Drone/Robot Obstacle Avoidance
- Drone Altitude Hold and Obstacle Avoidance
- Two-Wheeler Blind Spot Detection (BSD)

2 System Application Block Diagram

The MRS6241-P2840-M81 mmWave radar module requires external power supplies and provides 18 GPIOs. VCC is the main module supply, and VDDIO supplies the GPIOs and on-module peripherals. When the supply voltage does not exceed 3.6 V, VCC and VDDIO can be connected directly in the application.

V13 can be supplied by the internal VINT rail or an external low-power DC-DC converter.

The MRS6241-P2840-M81 mmWave radar module integrates CPUs for application development, so an external MCU is optional in the system design.

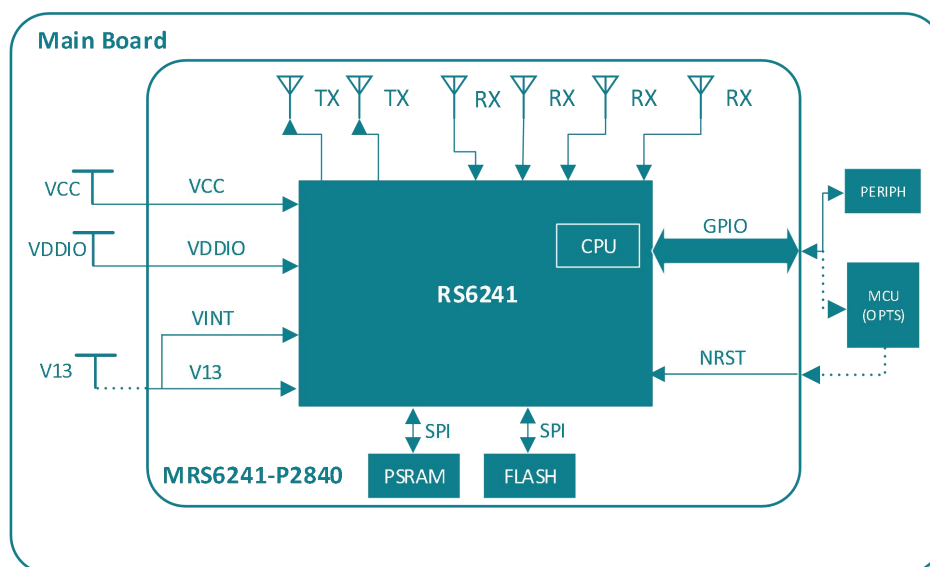


Figure 2-1 MRS6241-P2840-M81 System Application Block Diagram

3 Pin Definitions

3.1 Pin Definitions

The module pin layout is shown in the following figure.

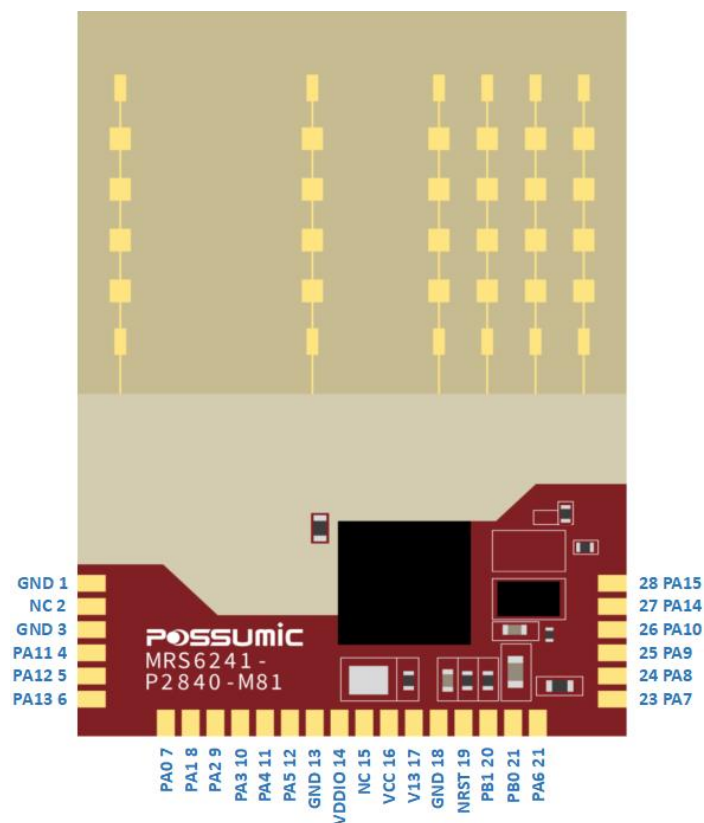


Figure 3-1 Pin Layout

3.2 Pin Descriptions

The module has 28 pins. Their functions are described in the following table.

Table 3-1 Module Pin Function Descriptions

Pin Name	No.	Type	Function
GND	1	G	Module ground
NC	2		
GND	3	G	Module ground
PA11	4	IO/WUP	PA11,WIO11,I2C1_SCL,CANFD_RX,PWM3,GPADC3
PA12	5	IO/WUP	PA12,WIO12,UART2_RX,UART1_RTS,PWM4,GPADC4
PA13	6	IO/WUP	PA13,WIO13,TEST,UART2_TX,MCO_CLK,PWM5,GPADC5
PA0	7	IO/WUP	PA0,WIO0,SPI0_CS,UART1_TX,CANFD_TX,PWM0
PA1	8	IO/WUP	PA1,WIO1,SPI0_CLK,UART1_RX,CANFD_RX,PWM1
PA2	9	IO/WUP	PA2,WIO2,SPI0_MOSI,UART1_RTS,I2C0_SDA,UART2_TX/LIN_TX,PWM2
PA3	10	IO/WUP	PA3,WIO3,SPI0_MISO,UART1_CTS,I2C0_SCL,UART2_RX/LIN_RX,PWM3

PA4	11	IO/WUP	PA4,WIO4,SPI0_HOLD,I2C0_SDA,PWM4
PA5	12	IO/WUP	PA5,WIO5,SPI0_WP,I2C0_SCL,PWM5
GND	13	G	Module ground
VDDIO	14	PI	VDDIO supplies the GPIOs and on-module peripherals, from 1.7 V to 3.6 V. When the supply voltage does not exceed 3.6 V, VCC and VDDIO can be connected directly in the application.
NC	15		
VCC	16	PI	Main module supply, 1.7 V to 5.5 V
V13	17	PI/PO	Module V13 supply. This rail is 1.3 V when an external DC-DC converter is used and measures 1.5 V when the internal LDO is used.
GND	18	G	Module ground
NRST	19	I	Radar-module reset signal
PB1	20	IO/WUP	PB1,WIO17,UART0_RX,PWM1
PB0	21	IO/WUP	PB0,WIO16,UART0_TX,PWM0
PA6	22	IO/WUP	PA6,WIO6,UART1_RTS,UART2_TX,I2C1_SDA,PWM6,GPADC6
PA7	23	IO/WUP	PA7,WIO7,UART1_CTS,UART2_RX,I2C1_SCL,PWM7,GPADC7
PA8	24	IO/WUP	PA8,WIO8,LXL1,UART1_TX,I2C1_SCL,PWM0,GPADC0
PA9	25	IO/WUP	PA9,WIO9,LXL2,UART1_RX,I2C1_SDA,PWM1,GPADC1
PA10	26	IO/WUP	PA10,WIO10,I2C1_SDA,CANFD_TX,PWM2,GPADC2
PA14	27	IO/WUP	PA14,WIO14,TWIO_SDA,UART2_RX,CANFD_TX,PWM6
PA15	28	IO/WUP	PA15,WIO15,TWIO_SCL,UART2_TX,CANFD_RX,PWM7

Note:

- **G** denotes ground.
- **PI** denotes a power input.
- **PO** denotes a power output.
- **GPIO** denotes a general-purpose digital pin.
- **I** denotes an input pin.
- **WUP** denotes a digital pin that can retain its state or wake the system in standby mode. WIOx denotes the sequential number of a WUP-type I/O.

4 Peripherals

4.1 GPIO

The MRS6241-P2840-M81 module provides 18 GPIOs, each with multiple multiplexed functions.

GPIO configuration features:

- Three modes: input, output, and high impedance
- Configurable input pull-up and pull-down
- Default output drive strength of 20 mA, configurable in 20 mA steps up to 80 mA
- I/O outputs support push-pull and open-drain modes
- External-interrupt I/Os support five trigger modes: rising edge, falling edge, both edges, high level, and low level
- The 18 wake-up I/Os can be configured to wake the system from standby through an external interrupt
- All I/Os can be configured to wake the system from idle through an external interrupt

Note:

Eight I/Os are multiplexed with eight ADC channels. A dedicated internal control signal selects between the digital and analog functions

4.2 UART

The MRS6241-P2840-M81 module has three embedded UART controllers: UART2 is used for debugging, while UART0 and UART1, which support automatic flow control, communicate with external devices. By default, UART0 is used for system boot configuration and firmware upgrades. The UARTs are compatible with the 16550 industry standard and support baud rates up to 3 Mbps. UART features include:

- Compliant with the industry-standard 16550 specification
- DMA controller interface support
- 64-byte receive and transmit FIFOs
- Programmable fractional baud rate
- Automatic hardware flow control
- Baud-rate detection
- RS-232 support
- Clock detection to wake the system from standby

4.3 I2C

The MRS6241-P2840-M81 module supports I2C0 and I2C1 communication and provides extensive pin multiplexing. I2C features include:

- Configurable as master or slave
- Standard mode (100 kbit/s)
- Fast mode (400 kbit/s)
- Fast-mode Plus (1,000 kbit/s)
- 7-bit and 10-bit addressing modes
- Multi-master systems
- 8-byte receive and transmit FIFO depth
- Programmable SDA hold time
- DMA support
- Address detection to wake the system from standby

4.4 PWM

The MRS6241-P2840-M81 module provides four PWM generators. Each generator outputs two PWM signals and one pair of complementary signals. Capture mode is available only on PWM channels 0, 2, 4, and 6. Capture detects rising and falling edges and calculates the capture-counter value for comparison with the match value.

PWM features include:

- Four PWM generators, eight PWM outputs, and four capture inputs (PWM0, PWM2, PWM4, and PWM6)
- Input capture and interrupt support
- Configurable dead time
- Pulse, periodic, and complementary signal outputs
- Adjustable duty cycle from 0% to 100%

4.5 SPI

The MRS6241-P2840-M81 module supports a four-wire SPI interface (QSPI). QSPI supports half-duplex, full-duplex, synchronous serial communication with peripherals. The interface can be configured as master or slave. Four combinations of serial-clock phase and polarity are supported. Features include:

- Clock frequency up to 96 MHz
- Motorola SPI mode
- Programmable master or slave mode
- Standard, dual, and quad SPI modes
- Programmable data width from 4 to 32 bits
- 32-word FIFO depth for both transmit and receive data
- Programmable sampling delay in master mode
- Programmable serial-clock polarity (SCPOL) and serial-clock phase (SCPH)
- Interrupt and DMA support
- Programmable instruction length, address length, and wait cycles in master mode
- Clock detection to wake the system from standby

4.6 GPADC

The MRS6241-P2840-M81 module includes a 10-channel analog switch and an asynchronous 12-bit SAR ADC with single-ended inputs. Channels 0 through 7 measure external input voltages, channel 8 measures VINT, and channel 9 is dedicated to measuring VCC.

GPADC features include:

- 12-bit resolution with 10 effective bits for the SAR A/D converter
- 10-channel multiplexer with eight general channels and two VCC voltage-detection channels
- Data FIFO depth up to 32
- DMA support
- Calibration and compensation support
- Internal VREF of 1.2 V with five input-voltage attenuation ratios: 2/3, 1/2, 1/3, 1/5, and 1/9. The maximum voltage is 3.6 V
- Maximum sampling frequency: 2 MHz
- Programmable oversampling: 16, 32, 64, 128
- 4,096× oversampling on one channel
- Comparison and data interrupts
- Four operating modes: single, single-cycle, continuous, and burst
- Hardware triggers (periodic internal-timer events or WIO input events) can start conversion in standby mode. When the data falls outside the comparison window, an interrupt is generated to wake the system.

4.7 CAN-FD

The CAN controller supports CAN 2.0B, CAN FD, and TTCAN. It includes a configurable receive buffer, two transmit buffers, and four acceptance filters.

The CAN controller supports the following features:

- CAN 2.0B: Payloads up to 8 bytes and data rates up to 1 Mbit/s
- CAN FD: Payloads up to 64 bytes; the data rate is limited by the transceiver and the CAN-CTRL core clock frequency
- TTCAN: Time-triggered CAN that can be used with CAN 2.0B or CAN FD
- Configurable receive-buffer size: three buffer slots with FIFO (first-in, first-out) behavior
- Transmit buffers: one primary transmit buffer and three secondary transmit buffers
- Four acceptance filters: the acceptance code can be checked bit by bit
- Extended functions: single-shot transmission mode and listen-only mode

4.8 LIN

The LIN controller supports the following features:

- Transmission rates up to 1 MHz
- Master mode initiates message transmission and reception for different frame IDs
- Slave mode responds to and processes messages with different IDs
- Arbitration adaptation for multiple slaves
- Adjustable break-field length
- Classic and enhanced checksum modes
- Configurable frame-slot widths in master mode

5 Hardware Operation

5.1 Hardware Startup

Supply the module with 3.3 V on VCC and 3.3 V on VDDIO; V13 is supplied internally.

Drive NRST low and then release it high. The hardware powers up and loads firmware from the external Flash into SRAM for execution.

5.2 Firmware Programming

Depending on the hardware configuration, either of the following methods can be used to place the chip in programming mode.

Method 1: The host controller can control the NRST pin through an I/O

- (1) Keep VCC powered and PB0 and PB2 high.
- (2) Drive NRST low and wait 20 ms.
- (3) Drive NRST high and wait 50 ms.
- (4) Drive PB0 (or PB2) low, and then drive NRST low.
- (5) Wait 20 ms, and then drive NRST high.
- (6) Wait 50 ms, and then drive PB0 (or PB2) high. The chip and Flash have now been reset and entered programming mode.

Method 2: For systems with an externally controlled power-off mechanism

- (1) Power off the Possumic chip and Flash or the module. If the external capacitance is 10 μ F, a 10 s power-off interval is recommended. Adjust the interval according to the external capacitance and the actual discharge rate.
- (2) Drive PB0 (or PB2) low.
- (3) Power the Possumic chip and Flash or the module back on, wait 50 ms, and drive PB0 (or PB2) high. The chip and Flash have now been reset and entered programming mode.

You can now use the host-side BurnTool to program firmware to Flash through any of the following interfaces: UART (PB0/PB1), SPI (PA0-PA3), or I2C (PA4/PA5).

6 Electrical Characteristics

6.1 Power-Supply Operating Conditions

Table 6-1 Module Power-Supply Operating Conditions

Name	Parameter Description	Minimum	Typical	Maximum	Unit
VCC	Module Power-Input Voltage	1.7	3.3	5.5	V
VDDIO	Module I/O Input Voltage	1.7	3.3	3.6	V
V13	Module Internal Supply Output Voltage	1.2	1.5	2.5	V
Ivcc	Current Supplied by the External Power Source	-	-	800	mA

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